Chapter 8: Markov Chains

it only matters where you are, not where you've been...

8.1 Introduction

A.A.Markov
1856-1922

So far, we have examined several stochastic processes using transition diagrams and First-Step Analysis.
The processes can be written as $\{X_0, X_1, X_2, \ldots\}$,
where X_t is the *state at time* t.

On the transition diagram, X_t corresponds to
which box we are in at step t.

In the Gambler's Ruin (Section 2.7), X_t is the amount of money the gambler possesses after toss t. In the model for gene spread (Section 3.7), X_t is the number of animals possessing the harmful allele A in generation t.

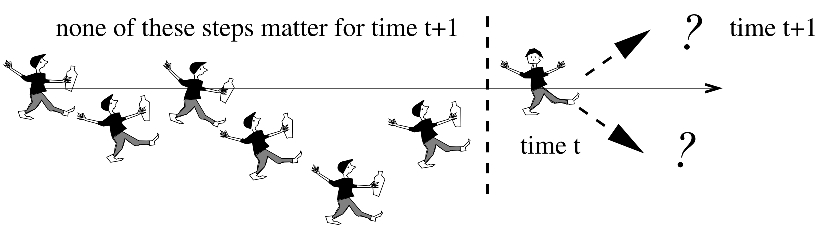
The processes that we have looked at via the transition diagram have a crucial property in common:

X_{t+1} *depends only on* X_t.

It does <u>not</u> depend upon $X_0, X_1, \ldots, X_{t-1}$.

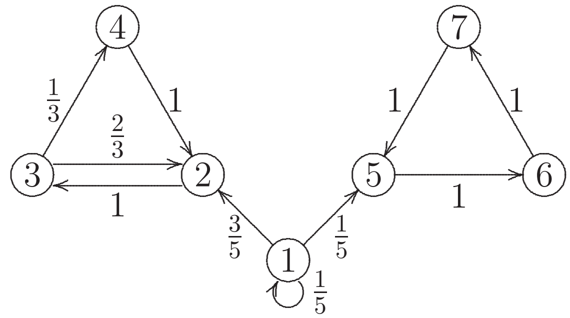
Processes like this are called *Markov Chains.*

Example: Random Walk (see Chapter 4)



In a Markov chain, the future depends only upon the present: NOT upon the past.

The text-book image of a Markov chain has a flea hopping about at random on the vertices of the transition diagram, according to the probabilities shown.



The transition diagram above shows a system with 7 possible states:

$$\textit{state space } S = \{1, 2, 3, 4, 5, 6, 7\}.$$

Questions of interest

- Starting from state 1, what is the probability of ever reaching state 7?
- Starting from state 2, what is the expected time taken to reach state 4?
- Starting from state 2, what is the long-run proportion of time spent in state 3?
- Starting from state 1, what is the probability of being in state 2 at time t? Does the probability converge as $t \to \infty$, and if so, to what?

We have been answering questions like the first two using first-step analysis since the start of STATS 325. In this chapter we develop a unified approach to all these questions using the matrix of transition probabilities, called the *transition matrix.*

8.2 Definitions

The Markov chain is the process $X_0, X_1, X_2, \ldots$.

Definition: The **state** of a Markov chain at time t is the *value of X_t.*

For example, if $X_t = 6$, we say *the process is in state 6 at time t.*

Definition: The **state space** of a Markov chain, S, is the set of values that each X_t can take. For example, $S = \{1, 2, 3, 4, 5, 6, 7\}$.

Let S have size N (possibly infinite).

Definition: A **trajectory** of a Markov chain is *a particular set of values for $X_0, X_1, X_2, \ldots$.*

For example, if $X_0 = 1$, $X_1 = 5$, and $X_2 = 6$, then the trajectory up to time $t = 2$ is $1, 5, 6$.

More generally, if we refer to the trajectory $s_0, s_1, s_2, s_3, \ldots$, we mean that $X_0 = s_0$, $X_1 = s_1$, $X_2 = s_2$, $X_3 = s_3$, ...

'Trajectory' is just a word meaning *'path'.*

Markov Property

The basic property of a Markov chain is that *only the most recent point in the trajectory affects what happens next.*

This is called the *Markov Property.*
It means that X_{t+1} *depends upon* X_t*, but it does not depend upon* $X_{t-1}, \ldots, X_1, X_0$.

We formulate the Markov Property in mathematical notation as follows:

$$\boxed{\mathbb{P}(X_{t+1} = s \mid X_t = s_t, X_{t-1} = s_{t-1}, \ldots, X_0 = s_0) \;=\; \mathbb{P}(X_{t+1} = s \mid X_t = s_t),}$$

for all $t = 1, 2, 3, \ldots$ and for all states $s_0, s_1, \ldots, s_t, s$.

Explanation:

$$\mathbb{P}(\underset{\substack{\uparrow \\ \textit{distribution} \\ \textit{of } X_{t+1}}}{X_{t+1} = s} \;\mid\; \underset{\substack{\uparrow \\ \textit{depends} \\ \textit{on } X_t}}{X_t = s_t}, \;\underbrace{\cancel{X_{t-1} = s_{t-1}, X_{t-2} = s_{t-2}, \ldots, X_1 = s_1, X_0 = s_0}}_{\substack{\uparrow \\ \textit{but whatever happened before time } t \\ \textit{doesn't matter.}}})$$

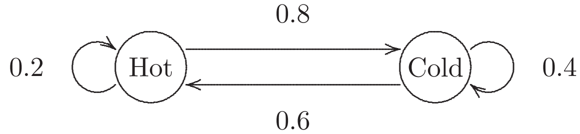
Definition: Let $\{X_0, X_1, X_2, \ldots\}$ be a sequence of discrete random variables. Then $\{X_0, X_1, X_2, \ldots\}$ is a **Markov chain** if *it satisfies the Markov property:*

$$\mathbb{P}(X_{t+1} = s \mid X_t = s_t, \ldots, X_0 = s_0) \;=\; \mathbb{P}(X_{t+1} = s \mid X_t = s_t),$$

for all $t = 1, 2, 3, \ldots$ *and for all states* $s_0, s_1, \ldots, s_t, s$.

8.3 The Transition Matrix

We have seen many examples of **transition diagrams** to describe Markov chains. The transition diagram is so-called because it shows the transitions between different states.



We can also summarize the probabilities in a **matrix:**

$$\begin{array}{cc} & \overbrace{\begin{array}{cc}\text{Hot} & \text{Cold}\end{array}}^{X_{t+1}} \\ X_t \left\{ \begin{array}{c}\text{Hot} \\ \text{Cold}\end{array} \right. & \begin{pmatrix} 0.2 & 0.8 \\ 0.6 & 0.4 \end{pmatrix} \end{array}$$

The matrix describing the Markov chain is called the ***transition matrix.*** It is the most important tool for analysing Markov chains.

Transition Matrix

$$\begin{array}{cc} & \overbrace{\longleftarrow \text{ list all states } \longrightarrow}^{X_{t+1}} \\ X_t \left\{ \begin{array}{c}\text{list}\\ \text{all}\\ \text{states}\end{array}\right. \updownarrow & \left(\begin{array}{c} \texttt{insert} \\ \texttt{probabilities} \\ p_{ij} \end{array} \right) \begin{array}{l} \longleftarrow \text{rows add to 1} \\ \longleftarrow \text{rows add to 1} \end{array} \end{array}$$

The transition matrix is usually given the symbol $P = (p_{ij})$.

In the transition matrix P:

- *the ROWS represent NOW, or FROM* (X_t)*;*
- *the COLUMNS represent NEXT, or TO* (X_{t+1})*;*
- *entry* (i, j) *is the CONDITIONAL probability that NEXT* $= j$*, given that NOW* $= i$*: the probability of going FROM state* i *TO state* j*.*

$$\boxed{p_{ij} = \mathbb{P}(X_{t+1} = j \mid X_t = i).}$$

Notes: 1. The transition matrix P must list *all* possible states in the state space S.

2. P is a *square matrix* $(N \times N)$, because X_{t+1} and X_t both take values in the same state space S (of size N).

3. The **rows** of P should each ***sum to 1:***

$$\sum_{j=1}^{N} p_{ij} = \sum_{j=1}^{N} \mathbb{P}(X_{t+1} = j \mid X_t = i) = \sum_{j=1}^{N} \mathbb{P}_{\{X_t = i\}}(X_{t+1} = j) = 1.$$

This simply states that X_{t+1} *must* take one of the listed values.

4. The **columns** of P do **not** in general sum to 1.

Definition: Let $\{X_0, X_1, X_2, \ldots\}$ be a Markov chain with state space S, where S has size N (possibly infinite). The **transition probabilities** of the Markov chain are

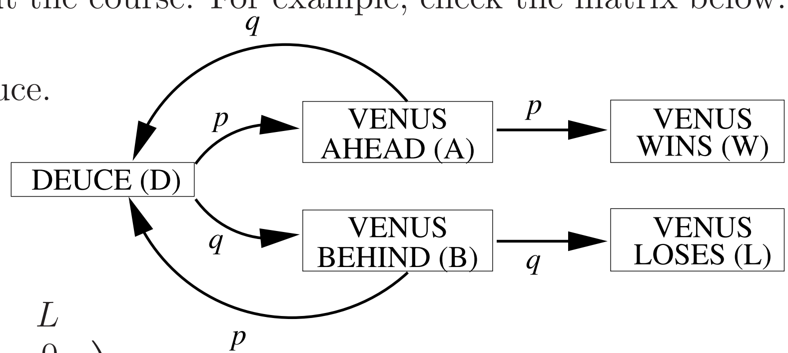
$$p_{ij} = \mathbb{P}(X_{t+1} = j \,|\, X_t = i) \quad \text{for } i, j \in S, \quad t = 0, 1, 2, \ldots$$

Definition: The **transition matrix** of the Markov chain is $P = (p_{ij})$.

8.4 Example: setting up the transition matrix

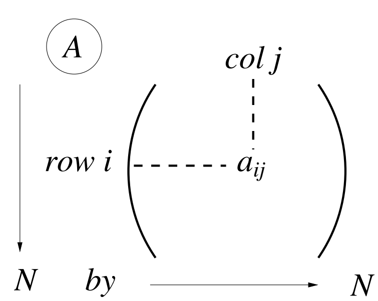
We can create a transition matrix for any of the transition diagrams we have seen in problems throughout the course. For example, check the matrix below.

Example: Tennis game at Deuce.



$$\begin{array}{c} \\ D \\ A \\ B \\ W \\ L \end{array}
\begin{array}{c} \begin{array}{ccccc} D & A & B & W & L \end{array} \\
\begin{pmatrix} 0 & p & q & 0 & 0 \\ q & 0 & 0 & p & 0 \\ p & 0 & 0 & 0 & q \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$

8.5 Matrix Revision



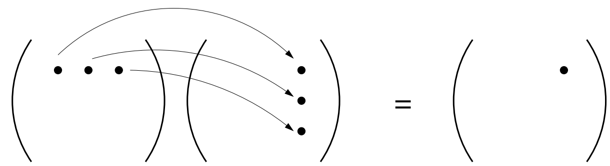
Notation

Let A be an $N \times N$ matrix.

We write $A = (a_{ij})$,
i.e. A comprises elements a_{ij}.

The (i, j) element of A is written both as a_{ij} and $(A)_{ij}$:
e.g. for matrix A^2 we might write $(A^2)_{ij}$.

Matrix multiplication



Let $A = (a_{ij})$ and $B = (b_{ij})$ be $N \times N$ matrices.

The product matrix is $A \times B = AB$, with elements $(AB)_{ij} = \sum_{k=1}^{N} a_{ik} b_{kj}$.

Summation notation for a matrix squared

Let A be an $N \times N$ matrix. Then

$$(A^2)_{ij} = \sum_{k=1}^{N} (A)_{ik}(A)_{kj} = \sum_{k=1}^{N} a_{ik} a_{kj}.$$

Pre-multiplication of a matrix by a vector

Let A be an $N \times N$ matrix, and let $\boldsymbol{\pi}$ be an $N \times 1$ column vector: $\boldsymbol{\pi} = \begin{pmatrix} \pi_1 \\ \vdots \\ \pi_N \end{pmatrix}$.

We can pre-multiply A by $\boldsymbol{\pi}^T$ to get a $1 \times N$ row vector, $\boldsymbol{\pi}^T A = \big((\boldsymbol{\pi}^T A)_1, \ldots, (\boldsymbol{\pi}^T A)_N\big)$, with elements

$$(\boldsymbol{\pi}^T A)_j = \sum_{i=1}^{N} \pi_i a_{ij}.$$

8.6 The t-step transition probabilities

Let $\{X_0, X_1, X_2, \ldots\}$ be a Markov chain with state space $S = \{1, 2, \ldots, N\}$.

Recall that the elements of the transition matrix P are defined as:

$$(P)_{ij} = p_{ij} = \mathbb{P}(X_1 = j \mid X_0 = i) = \mathbb{P}(X_{n+1} = j \mid X_n = i) \quad \text{for any } n.$$

p_{ij} is the probability of making a transition FROM state i TO state j in a SINGLE step.

Question: what is the probability of making a transition from state i to state j over **two** steps? *I.e. what is* $\mathbb{P}(X_2 = j \mid X_0 = i)$*?*

We are seeking $\mathbb{P}(X_2 = j \mid X_0 = i)$. Use the ***Partition Theorem:***

$$\begin{aligned}
\mathbb{P}(X_2 = j \mid X_0 = i) &= \mathbb{P}_i(X_2 = j) \quad \textit{(notation of Ch 2)} \\
&= \sum_{k=1}^{N} \mathbb{P}_i(X_2 = j \mid X_1 = k)\mathbb{P}_i(X_1 = k) \quad \textit{(Partition Thm)} \\
&= \sum_{k=1}^{N} \mathbb{P}(X_2 = j \mid X_1 = k,\ X_0 = i)\mathbb{P}(X_1 = k \mid X_0 = i) \\
&= \sum_{k=1}^{N} \mathbb{P}(X_2 = j \mid X_1 = k)\mathbb{P}(X_1 = k \mid X_0 = i) \\
&\qquad\qquad \textit{(Markov Property)} \\
&= \sum_{k=1}^{N} p_{kj} p_{ik} \quad \textit{(by definitions)} \\
&= \sum_{k=1}^{N} p_{ik} p_{kj} \quad \textit{(rearranging)} \\
&= (P^2)_{ij}. \quad \textit{(see Matrix Revision)}
\end{aligned}$$

The two-step transition probabilities are therefore given by ***the matrix*** P^2***:***

$$\boxed{\mathbb{P}(X_2 = j \mid X_0 = i) = \mathbb{P}(X_{n+2} = j \mid X_n = i) = \left(P^2\right)_{ij} \quad \text{for any } n.}$$

3-step transitions: We can find $\mathbb{P}(X_3 = j \mid X_0 = i)$ similarly, but conditioning on the state at time 2:

$$\begin{aligned}
\mathbb{P}(X_3 = j \mid X_0 = i) &= \sum_{k=1}^{N} \mathbb{P}(X_3 = j \mid X_2 = k)\mathbb{P}(X_2 = k \mid X_0 = i) \\
&= \sum_{k=1}^{N} p_{kj} \left(P^2\right)_{ik} \\
&= (P^3)_{ij}.
\end{aligned}$$

The three-step transition probabilities are therefore given by the matrix P^3:

$$\boxed{\mathbb{P}(X_3 = j \mid X_0 = i) = \mathbb{P}(X_{n+3} = j \mid X_n = i) = \left(P^3\right)_{ij} \quad \text{for any } n.}$$

General case: *t*-step transitions

The above working extends to show that the t-step transition probabilities are given by the matrix P^t for any t:

$$\boxed{\mathbb{P}(X_t = j \mid X_0 = i) = \mathbb{P}(X_{n+t} = j \mid X_n = i) = \left(P^t\right)_{ij} \quad \text{for any } n.}$$

We have proved the following Theorem.

Theorem 8.6: Let $\{X_0, X_1, X_2, \ldots\}$ be a Markov chain with $N \times N$ transition matrix P. Then the t-step transition probabilities are given by the matrix P^t. That is,

$$\mathbb{P}(X_t = j \mid X_0 = i) = \left(P^t\right)_{ij}.$$

It also follows that

$$\mathbb{P}(X_{n+t} = j \mid X_n = i) = \left(P^t\right)_{ij} \text{ for any } n. \qquad \square$$

8.7 Distribution of X_t

Let $\{X_0, X_1, X_2, \ldots\}$ be a Markov chain with state space $S = \{1, 2, \ldots, N\}$.

Now each X_t is a random variable, so it has a ***probability distribution.***

We can write the probability distribution of X_t as an $N \times 1$ ***vector.***

For example, consider X_0. Let $\boldsymbol{\pi}$ be an $N \times 1$ vector denoting the probability distribution of X_0:

$$\boldsymbol{\pi} = \begin{pmatrix} \pi_1 \\ \pi_2 \\ \vdots \\ \pi_N \end{pmatrix} = \begin{pmatrix} \mathbb{P}(X_0 = 1) \\ \mathbb{P}(X_0 = 2) \\ \vdots \\ \mathbb{P}(X_0 = N) \end{pmatrix}$$

In the flea model, this corresponds to *the flea choosing at random which vertex it starts off from at time 0, such that*

$$\mathbb{P}(\textit{flea chooses vertex } i \textit{ to start}) = \pi_i.$$

Notation: we will write $X_0 \sim \boldsymbol{\pi}^T$ to denote that the row vector of probabilities is given by the row vector $\boldsymbol{\pi}^T$.

Probability distribution of X_1

Use the Partition Rule, conditioning on X_0:

$$\begin{aligned}
\mathbb{P}(X_1 = j) &= \sum_{i=1}^{N} \mathbb{P}(X_1 = j \mid X_0 = i)\mathbb{P}(X_0 = i) \\
&= \sum_{i=1}^{N} p_{ij}\pi_i \quad \textit{by definitions} \\
&= \sum_{i=1}^{N} \pi_i p_{ij} \\
&= \left(\boldsymbol{\pi}^T P\right)_j .
\end{aligned}$$

(pre-multiplication by a vector from Section 8.5).

This shows that $\mathbb{P}(X_1 = j) = \left(\boldsymbol{\pi}^T P\right)_j$ *for all* j.
The row vector $\boldsymbol{\pi}^T P$ is therefore *the probability distribution of* X_1*:*

$$\boxed{\begin{array}{lcl} X_0 & \sim & \boldsymbol{\pi}^T \\ X_1 & \sim & \boldsymbol{\pi}^T P. \end{array}}$$

Probability distribution of X_2

Using the Partition Rule as before, conditioning again on X_0:

$$\mathbb{P}(X_2 = j) = \sum_{i=1}^{N} \mathbb{P}(X_2 = j \mid X_0 = i)\mathbb{P}(X_0 = i) = \sum_{i=1}^{N} \left(P^2\right)_{ij} \pi_i = \left(\boldsymbol{\pi}^T P^2\right)_j .$$

The row vector $\boldsymbol{\pi}^T P^2$ is therefore the probability distribution of X_2:

$$\boxed{\begin{array}{rcl} X_0 & \sim & \boldsymbol{\pi}^T \\ X_1 & \sim & \boldsymbol{\pi}^T P \\ X_2 & \sim & \boldsymbol{\pi}^T P^2 \\ & \vdots & \\ X_t & \sim & \boldsymbol{\pi}^T P^t. \end{array}}$$

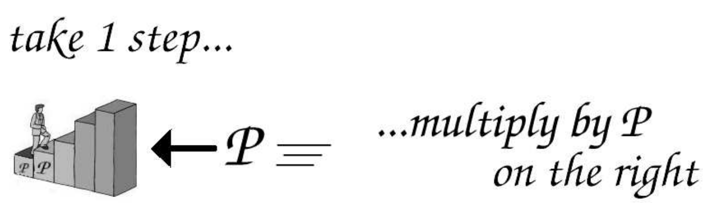
These results are summarized in the following Theorem.

Theorem 8.7: Let $\{X_0, X_1, X_2, \ldots\}$ be a Markov chain with $N \times N$ transition matrix P. If the probability distribution of X_0 is given by the $1 \times N$ row vector $\boldsymbol{\pi}^T$, then the probability distribution of X_t is given by the $1 \times N$ row vector $\boldsymbol{\pi}^T P^t$. That is,

$$X_0 \sim \boldsymbol{\pi}^T \quad \Rightarrow \quad X_t \sim \boldsymbol{\pi}^T P^t.$$

Note: The distribution of X_t is $X_t \sim \boldsymbol{\pi}^T P^t$.
The distribution of X_{t+1} is $X_{t+1} \sim \boldsymbol{\pi}^T P^{t+1}$.
Taking one step in the Markov chain corresponds to ***multiplying by P on the right.***

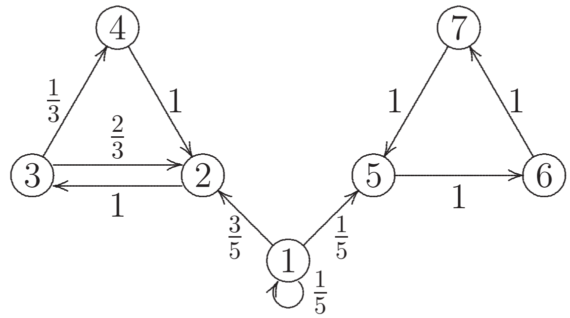
Note: The t-step transition matrix is P^t *(Theorem 8.6)*.
The $(t+1)$-step transition matrix is P^{t+1}.
Again, taking one step in the Markov chain corresponds to ***multiplying by P on the right.***



8.8 Trajectory Probability

Recall that a trajectory is a sequence of values for $X_0, X_1, \ldots, X_t$.

Because of the Markov Property, we can find the probability of any trajectory by multiplying together the starting probability and all subsequent single-step probabilities.



Example: Let $X_0 \sim (\frac{3}{4}, 0, \frac{1}{4}, 0, 0, 0, 0)$. What is the probability of the trajectory 1, 2, 3, 2, 3, 4?

$$\begin{aligned}
\mathbb{P}(1,2,3,2,3,4) &= \mathbb{P}(X_0 = 1) \times p_{12} \times p_{23} \times p_{32} \times p_{23} \times p_{34} \\
&= \tfrac{3}{4} \times \tfrac{3}{5} \times 1 \times \tfrac{2}{3} \times 1 \times \tfrac{1}{3} \\
&= \tfrac{1}{10}.
\end{aligned}$$

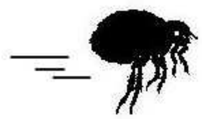
Proof in formal notation using the Markov Property:

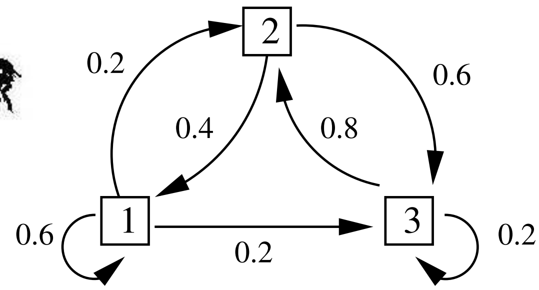
Let $X_0 \sim \boldsymbol{\pi}^T$. We wish to find the probability of the trajectory $s_0, s_1, s_2, \ldots, s_t$.

$$\begin{aligned}
&\mathbb{P}(X_0 = s_0, X_1 = s_1, \ldots, X_t = s_t) \\
&\quad = \mathbb{P}(X_t = s_t \,|\, X_{t-1} = s_{t-1}, \ldots, X_0 = s_0) \times \mathbb{P}(X_{t-1} = s_{t-1}, \ldots, X_0 = s_0) \\
&\quad = \mathbb{P}(X_t = s_t \,|\, X_{t-1} = s_{t-1}) \times \mathbb{P}(X_{t-1} = s_{t-1}, \ldots, X_0 = s_0) \quad \text{(Markov Property)} \\
&\quad = p_{s_{t-1}, s_t} \mathbb{P}(X_{t-1} = s_{t-1} \,|\, X_{t-2} = s_{t-2}, \ldots, X_0 = s_0) \times \mathbb{P}(X_{t-2} = s_{t-2}, \ldots, X_0 = s_0) \\
&\qquad \vdots \\
&\quad = p_{s_{t-1}, s_t} \times p_{s_{t-2}, s_{t-1}} \times \ldots \times p_{s_0, s_1} \times \mathbb{P}(X_0 = s_0) \\
&\quad = p_{s_{t-1}, s_t} \times p_{s_{t-2}, s_{t-1}} \times \ldots \times p_{s_0, s_1} \times \pi_{s_0}.
\end{aligned}$$

8.9 Worked Example: distribution of X_t and trajectory probabilities

Purpose-flea zooms around the vertices of the transition diagram opposite. Let X_t be Purpose-flea's state at time t $(t = 0, 1, \ldots)$.





(a) Find the transition matrix, P.

Answer: $P = \begin{pmatrix} 0.6 & 0.2 & 0.2 \\ 0.4 & 0 & 0.6 \\ 0 & 0.8 & 0.2 \end{pmatrix}$

(b) Find $\mathbb{P}(X_2 = 3 \mid X_0 = 1)$.

$$\begin{aligned} \mathbb{P}(X_2 = 3 \mid X_0 = 1) = \left(P^2\right)_{13} &= \begin{pmatrix} 0.6 & 0.2 & 0.2 \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix} \begin{pmatrix} \cdot & \cdot & 0.2 \\ \cdot & \cdot & 0.6 \\ \cdot & \cdot & 0.2 \end{pmatrix} \\ &= 0.6 \times 0.2 + 0.2 \times 0.6 + 0.2 \times 0.2 \\ &= 0.28. \end{aligned}$$

Note: we only need one element of the matrix P^2, so don't lose exam time by finding the whole matrix.

(c) Suppose that Purpose-flea is equally likely to start on any vertex at time 0. Find the probability distribution of X_1.

From this info, the distribution of X_0 is $\boldsymbol{\pi}^T = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$. We need $X_1 \sim \boldsymbol{\pi}^T P$.

$$\boldsymbol{\pi}^T P = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 0.6 & 0.2 & 0.2 \\ 0.4 & 0 & 0.6 \\ 0 & 0.8 & 0.2 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}.$$

Thus $X_1 \sim \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$ and therefore X_1 is also equally likely to be 1, 2, or 3.

(d) Suppose that Purpose-flea begins at vertex 1 at time 0. Find the probability distribution of X_2.

The distribution of X_0 *is now* $\boldsymbol{\pi}^T = (1, 0, 0)$. *We need* $X_2 \sim \boldsymbol{\pi}^T P^2$.

$$\boldsymbol{\pi}^T P^2 = (1 \;\; 0 \;\; 0) \begin{pmatrix} 0.6 & 0.2 & 0.2 \\ 0.4 & 0 & 0.6 \\ 0 & 0.8 & 0.2 \end{pmatrix} \begin{pmatrix} 0.6 & 0.2 & 0.2 \\ 0.4 & 0 & 0.6 \\ 0 & 0.8 & 0.2 \end{pmatrix}$$

$$= (0.6 \;\; 0.2 \;\; 0.2) \begin{pmatrix} 0.6 & 0.2 & 0.2 \\ 0.4 & 0 & 0.6 \\ 0 & 0.8 & 0.2 \end{pmatrix}$$

$$= (0.44 \quad 0.28 \quad 0.28).$$

Thus $\mathbb{P}(X_2 = 1) = 0.44$, $\mathbb{P}(X_2 = 2) = 0.28$, $\mathbb{P}(X_2 = 3) = 0.28$.

Note that it is quickest to multiply the vector by the matrix first: we don't need to compute P^2 *in entirety.*

(e) Suppose that Purpose-flea is equally likely to start on any vertex at time 0. Find the probability of obtaining the trajectory (3, 2, 1, 1, 3).

$$\begin{aligned} \mathbb{P}(3,2,1,1,3) &= \mathbb{P}(X_0 = 3) \times p_{32} \times p_{21} \times p_{11} \times p_{13} \quad \textit{(Section 8.8)} \\ &= \tfrac{1}{3} \times 0.8 \times 0.4 \times 0.6 \times 0.2 \\ &= 0.0128. \end{aligned}$$